Data Analytics Insights Report: I-94 Westbound Traffic

Prepared by: Kyi Htoo Aung | Smart City Traffic Intelligence Capstone, Part 1 | For the Smart City Mobility Team | Data: 48,204 hourly records, 2 Oct 2012 to 30 Sep 2018 (SQLite, Python/SciPy and Power BI)

Key message

Congestion on this corridor is driven by **time of day and day type, not weather**. On weekdays, average volume exceeds the 5,500 congestion threshold at 07:00 to 08:00 and 15:00 to 17:00 (peaking at about 6,300 vehicles per hour at 16:00 in 2017), while the apparent weather effects largely disappear once the hour is taken into account. Raw annual totals are also misleading because data coverage varies widely between years, so the team should report coverage-adjusted measures and invest in sensor data quality.

1. Annual traffic trends (SQL, Task 1.2)

Year	Total volume	Change vs prior	% change	Hours covered	Avg vehicles/hour	Avg/hour % change
2012	8,208,767			24%	3,227	
2013	28,177,412	+19,968,645	+243.3%	83%	3,310	+2.6%
2014	15,731,289	-12,446,123	-44.2%	51%	3,270	-1.2%
2015	14,181,206	-1,550,083	-9.9%	41%	3,258	-0.4%
2016	29,494,821	+15,313,615	+108.0%	89%	3,194	-2.0%
2017	35,428,156	+5,933,335	+20.1%	100%	3,377	+5.7%

Observation 1: raw totals track data coverage, not demand. Totals rose in 2013 (+243%), fell in 2014 (-44%) and 2015 (-10%), then rose in 2016 (+108%) and 2017 (+20%). These swings mirror coverage: 2012 starts in October, and no data exists from 8 Aug 2014 to 11 Jun 2015, so 2014 and 2015 each hold only about half a year. Repeated timestamps (7,629 rows) inflate totals further.

Observation 2: underlying demand was flat, then grew in 2017. Averaging one record per hour, volume held steady at about 3,190 to 3,310 vehicles per hour from 2012 to 2016 (2016 lowest), then rose 5.7% to 3,377 in 2017, the only year with near-complete coverage (99.5%). A like-for-like October to December comparison confirms this: 3,322 per hour in 2017 against 3,087 in 2016 (+7.6%). 2017 is the most reliable baseline year for planning.

2. Temperature around holidays (SQL, Task 1.3)

Holiday	Date	Avg temp	Min / max	YoY temp change	Avg traffic/hour	vs comparable day
New Year's Day	2016-01-01	-6.1 °C	-9.2 / -1.2 °C		1,832	-48.4%
New Year's Day	2017-01-02	-1.3 °C	-2.7 / -0.1 °C	+4.8 °C	2,091	-35.5%
Labor Day	2015-09-07	22.2 °C	18.7 / 26.5 °C		2,430	-34.5%
Labor Day	2016-09-05	21.9 °C	18.7 / 24.8 °C	-0.3 °C	2,152	-34.7%
Labor Day	2017-09-04	18.1 °C	11.8 / 23.1 °C	-3.8 °C	2,395	-32.7%

New Year's Day 2015 is absent (it falls in the data gap), and the 2017 holiday was observed on Monday 2 January. New Year's Day 2017 was 4.8 °C milder than 2016, and Labor Day cooled each year (22.2, 21.9 and 18.1 °C). **These changes do not appear to drive traffic:** every holiday ran 33% to 48% below comparable normal days, and Labor Day 2017 carried almost as much traffic as 2015 despite being 4.1 °C cooler, while the wet 2016 Labor Day (rain and thunderstorms) carried the least. The holiday calendar and precipitation matter far more than temperature (only five holiday days, so this is indicative).

3. Descriptive statistics and correlation (Task 2)

Hourly volume has a **mean of 3,260**, **median of 3,380**, **standard deviation of 1,987**, **variance of 3,947,615** and a **range of 7,280** (0 to 7,280). Variability is very high (coefficient of variation 61%) and the distribution is bimodal: a cluster of quiet night hours (Q1 = 1,193) and a cluster of busy daytime hours (Q3 = 4,933). The mean sits slightly below the median because night hours pull it down, so a single average describes a 'typical hour' poorly and planning should use hour-specific figures.

Temperature and volume have a **weak positive** correlation (Pearson r = 0.13 after removing ten 0 K sensor errors; Spearman rho = 0.13). Warmer hours tend to be slightly busier, but temperature explains under 2% of the variation (R squared = 0.018). The result is statistically significant only because n is large. **Correlation does not imply causation:** temperature rises during the day and in summer, exactly when commuting and leisure travel peak, so time of day and season act as confounders. Warm weather does not itself put cars on the road.

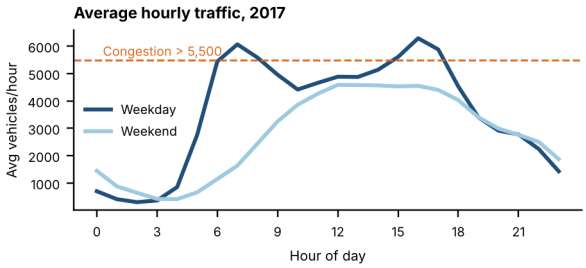
4. Probability and congestion (Task 3; congestion = volume above 5,500)

Measure	Value	Measure	Value
P(Congestion)	0.1473	P(Clear | Congestion)	0.2483
P(Clear Weather)	0.2778	P(High Temp > 292 K | Congestion)	0.2630
P(Congestion AND Clear)	0.0366	P(Congestion) x P(Clear)	0.0409
Odds ratio, Clear vs Cloudy	0.735 (95% CI 0.69 to 0.79)	Odds ratio adjusted for hour	1.006

About 1 hour in 7 is congested (14.7%). **Independence:** P(Congestion AND Clear) = 0.0366 is below P(C) x P(Clear) = 0.0409, so the events are not strictly independent (chi-square $p < 0.001$), but the gap is small (observed is 0.89 of expected). Congestion is somewhat less likely in clear than cloudy weather (odds ratio 0.735). However, clear skies make up 33% of night hours but only 23% of peak hours; once we compare clear and cloudy hours *at the same hour of day*, the odds ratio is 1.006, meaning no real difference. High temperature raises the chance slightly (26.3% of congested hours vs 24.4% of all hours). **Conclusion:** weather is only weakly associated with congestion and most of that association is explained by when different weather occurs.

5. Power BI dashboard findings (Task 4)

The dashboard (40,575 unique hours; average 3,260 vehicles per hour; average 8.1 °C) shows a classic commuter profile: traffic is lowest at 03:00 (about 380), peaks at 07:00 (4,880) and is highest at **16:00 (5,800)**. Split by day type (chart), weekdays exceed the threshold at 07:00 to 08:00 and 15:00 to 17:00 (about 6,070 and 6,290 at the two peaks), whereas weekends never do (maximum about 4,590 at midday) and lack the morning peak. By weather, Clouds has the highest average (3,618) and Squall the lowest (2,062), a gap of 1,556; but Squall has only 4 records, so Fog (2,704, 912 records) is the more reliable low. The scatter plot shows higher traffic mainly above 25 °C (4,404 to 4,781 per hour), with outliers from 0 K sensor errors and a near-zero traffic weekend on 23 to 24 July 2016 (likely a closure or sensor fault).



6. Implications for the Smart City Mobility Team

- **Target the two weekday peaks (07:00 to 08:00 and 15:00 to 17:00).** Between 67% and 89% of these hours are congested, so ramp metering, incident response and staggered work-hour campaigns should focus here rather than on weather events; the 16:00 peak is the most severe.
- **Use time-of-day as the primary planning variable.** Weather-triggered measures alone would be poorly targeted; any weather alerting should be layered on hour-of-day and day-type baselines (to be modelled in Parts 2 and 3).
- **Schedule roadworks on holidays and overnight.** Holidays run 33% to 48% below normal and 01:00 to 04:00 averages under 750 vehicles per hour, offering low-disruption maintenance windows.
- **Fix data quality before reporting trends.** Report coverage-adjusted KPIs (average per hour or day), de-duplicate timestamps, investigate the 2014 to 2015 gap and flag impossible sensor values; these rules feed directly into the Part 2 cleaning pipeline.

7. Limitations

The data covers a single westbound sensor location, so findings may not generalise to other corridors or directions. Coverage gaps (late 2012 start, the 2014 to 2015 gap, partial 2018) and repeated timestamps limit year-on-year comparisons. Weather labels are one category per record, and the 5,500 threshold is a fixed rule rather than a capacity-based measure, so congestion probabilities depend on that choice. All relationships reported here are associations, not causal effects; Part 3 will test them in multivariate models that control for time, day type and season.